

Safety Data Sheet

According to Notification of Thailand Ministry of Industry B.E. 2555 (2012)



PETRONAS HIDRAULIK 10W

Version 0.0

Release Date: 2019/07/01

1. PRODUCT AND COMPANY IDENTIFICATION

Product name	: PETRONAS HIDRAULIK 10W
Other means of identification	: None
Recommended use	: Lubricating oil
Limitation of use	: This product should not be used for purposes other than those specified without the advice of an expert
Supplier	: PETRONAS International Marketing (Thailand) Co.,Ltd
Registered corporate address	: 388 Exchange Tower, Unit Nos. 4101-4104 41 st floor, Sukhumvit Road, Kweang Klongtoey, Khet Klongtoey, Bangkok 10110
Emergency phone numbers	: +662-663 5881-5

2. HAZARDS IDENTIFICATION

GHS Classification

Skin corrosion/irritation	: Category 3
Serious eye damage/eye irritation	: Category 2B

GHS Label element

Hazard pictograms	: No
Signal word	: Warning
Hazard statements	: H316 Causes mild skin irritation. H319 Causes serious eye irritation.
Precautionary statements	: Prevention: P261: Avoid breathing vapours P264: Wash thoroughly after handling P271: Use only outdoors or in a well-ventilated area Response: P304+P340: If inhaled: Remove victim to fresh air and keep at rest in a position comfortable for breathing P305+P351+P338: If in eye: Rinse cautiously with water for several minutes, Remove contact lenses, if present and easy to do. P332+P313: If on skin, wash with plenty of soap and water Disposal: Store in a well-ventilated place. Keep container tightly closed

Other hazards which do not result in classification

No data available.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

Hazardous components

Chemical Name	CAS-No.	Concentration (%)
Distillates (petroleum), solvent- dewaxed heavy paraffinic	64742-65-0	90 - 95%
Additive	Proprietary	1 - 5%

4. FIRST AID MEASURES

If inhaled	: If unconscious, place in recovery position and seek medical advice. If symptoms persist, call a medical doctor.
In case of skin contact	: If on skin, rinse well with water. If on clothes, remove clothes. If skin irritation persists, call a medical doctor.
In case of eye contact	: Remove contact lenses. Protect unharmed eye. Keep eye wide open while rinsing. If eye irritation persists, consult a specialist. Immediately flush eye(s) with plenty of water.
If swallowed	: Keep respiratory tract clear. Do NOT give milk or alcoholic beverages. Never give anything by mouth to an unconscious person. Take victim immediately to hospital. If symptoms persist, call a medical doctor. Induce vomiting immediately and call a medical doctor.
General advice	: Move out of dangerous area. Show this safety data sheet to the doctor in attendance. Do not leave the victim unattended.
Most important symptoms and effects, both acute and delayed	: No information available.

5. FIREFIGHTING MEASURES

Unsuitable media	extinguishing	: High volume water jet
Specific hazards during firefighting		: Do not allow run-off from firefighting to enter drains or water courses.
Specific methods	extinguishing	: Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.
Special protective equipment for firefighters		: Wear self-contained breathing apparatus for firefighting if necessary.



6. ACCIDENTAL RELEASE MEASURES

- Personal precautions, protective equipment and emergency procedures : Use personal protective equipment.
- Environmental precautions : If the product contaminates rivers and lakes or drains inform respective authorities.
Prevent product from entering drains.
Prevent further leakage or spillage if safe to do so.
- Methods and materials for containment and cleaning up : Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust).
Keep in suitable, closed containers for disposal.

7. HANDLING AND STORAGE

- Advice on protection against fire and explosion : Normal measures for preventive fire protection.
- Advice on safe handling : Do not breathe vapours/dust.
Avoid exposure - obtain special instructions before use.
For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.
Dispose of rinse water in accordance with local and national regulations.
- Conditions for safe storage : Avoid contact with skin and eyes.
Keep container tightly closed in a dry and well-ventilated place.
Observe label precautions.
Electrical installations / working materials must comply with the technological safety standards.
Containers which are opened must be carefully resealed and kept upright to prevent leakage.

8. EXPOSURE CONTROLS/PERSONAL

PROTECTION Components with workplace

control parameters

Mineral Oil 64742-65-0	ACGIH TLV-TWA	5 mg/m3
	ACGIH TLV-STEL	10 mg/m3
	OSHA PEL-TWA	5 mg/m3

Personal protective equipment

Respiratory protection	: In the case of vapour formation use a respirator with an approved filter.
Hand protection	
Remarks	: The suitability for a specific workplace should be discussed with the producers of the protective gloves.
Eye protection	: Eye wash bottle with pure water. Tightly fitting safety goggles. Wear face-shield and protective suit for abnormal processing problems.
Skin and body protection	: Impervious clothing. Choose body protection according to the amount and concentration of the dangerous substance at the work place.
Hygiene measures	: When using do not eat or drink. When using do not smoke. Wash hands before breaks and at the end of workday.

9. PHYSICAL AND CHEMICAL PROPERTIES

a) Appearance	Bright&Clear
b) Odour	Characteristic
c) Odour Threshold	No data available
d) pH	No data available
e) Melting point/freezing point	No data available
f) Initial boiling point and boiling range	No data available
g) Flash point	252°C
h) Evaporation rate	No data available
i) Flammability (solid, gas)	No data available
j) Upper/lower flammability or explosive limits	No data available
k) Vapour pressure	No data available
l) Vapour density	No data available
m) Relative density	No data available
n) Water solubility	Insoluble
o) Partition coefficient: noctanol/water log Pow	No data available
p) Auto ignition temperature	No data available
q) Decomposition temperature	No data available
r) Viscosity	44 mm ² /s @40°C

10. STABILITY AND REACTIVITY

Reactivity	: Hazardous polymerisation does not occur.
Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: No dangerous reaction known under conditions of normal use.
Conditions to avoid	: Heat, sparks, flame and build-up of static electricity.
Incompatible materials	: Strong oxidizing agents.
Hazardous decomposition products	: Fumes, smoke, carbon monoxide

11. TOXICOLOGICAL INFORMATION**Information on the likely routes of exposure**

Inhalation : Prolonged breathing of vapors can cause headache, dizziness, nausea, respiratory irritation or chemical pneumonitis.

Skin contact : Slightly irritating.

Eye contact : Slightly Irritating.

Ingestion : Can cause stomach ache and vomiting.

Symptoms related to the physical, chemical and toxicological characteristics;

Main hazard, if ingested, is aspiration into the lungs and subsequent pneumonitis. Heating can generate vapors that may cause respiratory irritation, nausea and headaches. Inhalation hazard at room temperature is unlikely due to the low volatility of this product.

Delayed and immediate effects and also chronic effects from short and long term exposure;**Immediate effects**

May cause respiratory irritation, headache, nausea. Mildly irritating to skin and eyes.

Chronic effects :

Prolonged and repeated contact with skin can cause deflating and drying of the skin resulting in skin irritation and dermatitis.

Numerical measures of toxicity

Components	Acute toxicity
Distillates (petroleum), solvent-dewaxed heavy paraffinic	LD50 (oral rat) :> 5000 mg/kg LD50 (Dermal rabbit) :> 5000 mg/kg LC50 (Inhalation rat) :>2.4 mg/l

Classification of Health Hazards

Acute oral toxicity estimate	Not classified
Acute dermal toxicity estimate	Not classified
Acute inhalation toxicity estimate	Not classified
Skin corrosion / irritation	Category 3
Serious eye damage/eye irritation	Category 2B
Respiratory or skin sensitization	Not sensitization

Germ cell mutagenicity

Not classified

Highly refined mineral oils are not classified as carcinogenic by the International Agency for Research on Cancer (IARC)

Reproductive toxicity	No data available
Specific target organ toxicity - single exposure	No data available
Specific target organ toxicity - repeated exposure	No data available
Aspiration hazard	No data available



12. ECOLOGICAL INFORMATION

Ecotoxicity

Components	Result
Distillates (petroleum), solvent-dewaxed heavy paraffinic	LC50 Fish (<i>Salmo gairdneri</i>) > 1000 mg/l (96 h) EC50 Crustacea (<i>Daphnia magna</i>) > 1000 mg/l (48 h) ErC50 Algae (<i>Scenedesmus subspicatus</i>) > 1000 mg/l (96h) NOEC Fish (<i>Pimephales promelas</i>) > 5000 mg/l (7 days) NOEC Crustacea (<i>Ceriodaphnia</i> sp.) 552 mg/l (7 days)

Acute hazards to the aquatic environment estimate : Not classified
Long-term hazards to the aquatic environment estimate : Not classified

Persistence and degradability	No data available
Bio accumulative potential	No data available
Mobility in soil	Move under natural forces to the groundwater
Other adverse effects	Long term effect to the aquatic environment.

13. DISPOSAL CONSIDERATIONS

Disposal methods

Waste from residues	: Do not dispose of waste into sewer. Do not contaminate ponds, waterways or ditches with chemical or used container. Send to a licensed waste management company. The product should not be allowed to enter drains, water courses or the soil.
Contaminated packaging	: Empty remaining contents. Dispose of as unused product. Do not re-use empty containers.

14. TRANSPORT INFORMATION

UN number	No data available
UN proper shipping name	No data available
Transport hazard class(es)	No regulated
Packaging group	No data available

PETRONAS HIDRAULIK 10W



Version 0.0

Release Date: 2019/07/01

Environmental hazards	No data available
Transport in bulk	No data available
Special precautions for user	No data available

15. REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance
or mixture GLOBAL INVENTORIES

component	USA (TS CA)	EU (EINE CS)	AU (AI CS)	JAP (EN CS)	KOR (ELCI)	CHN (IEC SC)	PHLP (PIC CS)	CAN (DSL/ NDSL)	NZ (NZI OC)
	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

16. OTHER INFORMATION

Revision Date : 2019/07/01

Full text of other abbreviations

(Q)SAR	-	(Quantitative) Structure Activity Relationship
ACGIH	-	American Conference of Governmental Industrial Hygienists
AICS	-	Australian Inventory of Chemical Substances
ANTT	-	National Agency for Transport by Land of Brazil
ASTM	-	American Society for the Testing of Materials
bw	-	Body weight
CCHC	-	Chemicals Classification and Hazard Communication
CMR	-	Carcinogen, Mutagen or Reproductive Toxicant
CPR	-	Controlled Products Regulations
DIN	-	Standard of the German Institute for Standardisation
DSL	-	Domestic Substances List (Canada)
ECx	-	Concentration associated with x% response
ELx	-	Loading rate associated with x% response
EmS	-	Emergency Schedule
ENCS	-	Existing and New Chemical Substances (Japan)
ErCx	-	Concentration associated with x% growth rate response
GHS	-	Globally Harmonized System
GLP	-	Good Laboratory Practice
IARC	-	International Agency for Research on Cancer
IATA	-	International Air Transport Association
IBC	-	International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk
IC50	-	Half maximal inhibitory concentration
ICAO	-	International Civil Aviation Organization
ICOP	-	Industry Code of Practice on Chemicals Classification and Hazard Communication

PETRONAS HIDRAULIK 10W



Version 0.0

Release Date: 2019/07/01

IECSC	-	Inventory of Existing Chemical Substances in China
IMDG	-	International Maritime Dangerous Goods
IMO	-	International Maritime Organization
ISHL	-	Industrial Safety and Health Law (Japan)
ISO	-	International Organisation for Standardization
KECI	-	Korea Existing Chemicals Inventory
LC50	-	Lethal Concentration to 50 % of a test population
LD50	-	Lethal Dose to 50% of a test population (Median Lethal Dose)
MARPOL	-	International Convention for the Prevention of Pollution from Ships
MY PEL	-	Malaysian Permissible Exposure Limit
n.o.s.	-	Not Otherwise Specified
Nch	-	Chilean Norm
NITE	-	National Institute of Technology and Evaluation
NO(A)EC	-	No Observed (Adverse) Effect Concentration
NO(A)EL	-	No Observed (Adverse) Effect Level
NOELR	-	No Observable Effect Loading Rate
NOM	-	Official Mexican Norm
NTP	-	National Toxicology Program
NZIoC	-	New Zealand Inventory of Chemicals
OCSP	-	Office of Chemical Safety and Pollution Prevention
OECD	-	Organization for Economic Co-operation and Development
PBT	-	Persistent, Bioaccumulative and Toxic
PICCS	-	Philippines Inventory of Chemicals and Chemical Substances
REACH	-	Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals
SADT	-	Self-Accelerating Decomposition Temperature
SDS	-	Safety Data Sheet
STEL	-	Short Term Exposure Limit
TCSI	-	Taiwan Chemical Substance Inventory
TDG	-	Transportation of Dangerous Goods
TSCA	-	Toxic Substances Control Act (United States)
TWA	-	Time Weighted Average
UN	-	United Nations
UNRTDG	-	United Nations Recommendations on the Transport of Dangerous Goods
UVCB	-	Unknown or Variable Composition, Complex Reaction Products and Biological Materials
vPvB	-	Very Persistent and Very Bioaccumulative
WHMIS	-	Workplace Hazardous Materials Information System



Disclaimer

To the best of PETRONAS knowledge, the information contained herein is accurate and reliable as of the date of publication. Any use of information provided should be undertaken in consultation with a professional with the appropriate expertise, as necessary in the circumstances. No warranty, express or implied, is given as to the quality, accuracy, reliability, applicability or completeness of the contents of this SDS. The information presented here pertains only to the product as shipped. It is the responsibility of the customer to ensure that any activities relating to the product comply with all federal, state or local laws. Any hazards associated with any product regulatory requirements are subject to change and may differ between various locations. Except to the extent required by law, re-publication or retransmission of this SDS, in whole or in part, is strictly prohibited. PETRONAS does not take responsibility for use, transportation, storage, handling, packaging or disposal of the product that is beyond our knowledge and recommendation by this SDS.

Product Stewardship Advisory:

PETRONAS aims to increase awareness of all the hazards associated with the storage, handling and use of its products. Thoroughly reviewing the accompanying Safety Data Sheets and disseminating the information to all dependent and interested parties is an essential part of any 'Responsible Care' program.

TH / EN